WEAPONS SYSTEMS

AIR-TO-AIR MISSILES

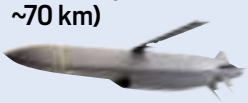
» Meteor (Ramjet-powered, >100 km)

» MICA (Medium-range; infrared & radar-guided variants)



AIR-TO-SURFACE MISSILES

» AM39 Exocet (Anti-ship missile, ~70 km)



» SCALP (Long-range cruise missile, >560 km)

PRECISION-GUIDED MUNITIONS



» Hammer (Navigation system-guided bombs with rocket booster)

» Paveway (Laser-guided bombs)

WHAT IT IS UP AGAINST

The induction of Rafale-Ms will redefine India's strategic reach against its key adversaries

CHINA

↘ People's Liberation Army Navy operates around 600 aircraft, comprising both fixed-wing jets and helicopters

↘ **SHENYANG J-15**
Primary carrier-based 4.5-gen multirole fighters, also known as Flying Shark;
59 in service



↘ **SHENYANG J-11**
Air superiority 4th-gen fighters;
50 in service

↘ The fleet also features J-15D electronic warfare variants and J-15T models equipped for catapult-assisted takeoff

PAKISTAN

↘ India's other key adversary has no aircraft carrier, and thus no naval jet

↘ Rafale-Ms will ensure superiority over the Arabian Sea against the Pakistan Air Force's JF-17 and F-16 jets besides striking capabilities against its naval frigates/submarines

Vishal, also ensures operational flexibility.

With their smaller size and lighter weight, marine fighter jets maximise efficiency while utilising the limited space available on aircraft carriers. Experts claim that, unlike their air force counterparts, naval jets possess specialised avionics, sensors and communication equipment for maritime operations. Built to withstand the challenges of saltwater environments, they have corrosion-resistant materials and coatings.

The Rafale-M has an exceptionally strong airframe and a reinforced undercarriage for carrier operations. It is a twin-engine, single-seat aircraft with folding wings that is capable of undertaking 'quick reaction alert', air defence and air policing missions, nuclear deterrence duties, power projection, deep strike missions, air support for ground forces, reconnaissance and pilot training sorties. Adept at both air-to-ground strikes and air-to-air attacks and interceptions during the same sortie, the service ceiling of the Rafale-M—the altitude at which an aircraft can no longer rise faster than 100 feet per minute under standard air conditions—stands at 50,000 feet.

According to Dassault, the Rafale-M has the RBE2-AA electronic scanning radar and Front Sector Optronics (FSO) to detect and track targets, while the SPECTRA internal 'electronic warfare' (EW) system enables long-range detection and identification of hostile radars, missiles and lasers. Rafale-Ms' unique data fusion capability yields a simplified and consolidated picture to the pilot. All these helped it beat the other naval fighter in contention for the Indian Navy's multi-role carrier borne fighters (MRCBF) programme—the US-made Boeing F/A-18 Super Hornet.

The need for a platform like the Rafale-M was also felt urgently due to issues with the 40-45 fourth-generation MiG 29K jets that are operated from INS *Vikramaditya*. The jets—shore-based aircraft modified for the navy—and associated equipment were ordered for $2.2 billion (Rs 18,750 crore, at current exchange rate) in two tranches in 2004 and 2010 from Russia. But the MiGs have a terrible record of accidents, with multiple crashes making them not entirely reliable. The Comptroller and Auditor General of India (CAG), in a report sometime back, had also flagged issues with the MiG-29K's airframe, its RD MK-33 engine and its fly-by-wire system. Moreover, the Russians are still heavily engaged in a conflict with Ukraine, and ceased to be reliable suppliers of spares and new platforms.

Compared to the MiG 29K, the Rafale is a vastly superior machine. "Its sensors and weapons are certainly technologically superior, and it will provide much more punch to the Indian Navy's striking abilities. Moreover, the Rafale-M is a proven aircraft, and even in small numbers, it will provide additional sorties as its serviceability is very good," says Admiral Prakash.

However, since INS *Vikrant* was designed for the LCA Navy (the naval version of the Light Combat Aircraft) and MiG 29K, the shipbuilder has to make some changes according to its requirements. Currently, the LCA Navy is superseded by the Twin Engine Deck Based Fighter (TEDBF), which is being developed by the Aeronautical Development Agency (ADA) for the navy. Its first test flight is expected by 2026 and induction by 2031. Till the time this indigenous TEDBF is ready and gets inducted in good numbers for long-term operational sovereignty, the Rafale-Ms will be the Indian Navy's aerial workhorse. ■